ZK-TRUST LEDGER

Problem

1. **Blockchain Transparency Challenge:**
   - "Blockchain's transparent ledger compromises user privacy by allowing traceability of all transactions."
2. **Privacy Concerns:**
   - "Advanced tools can link blockchain addresses to real identities, eroding the anonymity blockchain promises."
3. **Regulatory Compliance:**
   - "Regulators demand transparency for anti-money laundering (AML) and counter-terrorism financing (CTF), conflicting with privacy needs."
4. **Technical Dilemma:**
   - **Formula:** Privacy Risk = f(Transaction, Traceability)
   - **Formula:** Compliance Need=f(Regulatory, Transparency)
5. **The Need for Balance:**
   - "A solution is required that optimizes privacy (P) while satisfying compliance (C), where Balance=$f(P,C)$."

PREREQUISITES TO UNDERSTAND ASSOCIATION SETS:

1. **Blockchain Transparency**: Public blockchains are inherently transparent, meaning all transactions are publicly visible and traceable. This transparency can compromise user privacy.

2. **Pseudonymity and its Limitations**: Traditional blockchain transactions offer pseudonymity (users are represented by addresses, not personal identities) but advanced analysis tools can often de-anonymize these addresses.

3. **Zero-Knowledge Proofs (ZK-SNARKs)**: A technology that allows a user to prove the possession of certain information without revealing the information itself. It has two key properties:
   - **Zero-Knowledge**: The proof reveals nothing other than the fact that the statement is true.
   - **Succinctness**: The proofs are small in size and quick to verify.

4. **Privacy-Enhancing Protocols**: Systems like Tornado Cash that allow users to interact with the blockchain in ways that obscure the links between specific transactions, enhancing privacy.

UNDERSTANDING ASSOCIATION SETS:

- **Definition:** An association set is a collection of blockchain transactions (or transaction inputs) that a user can prove their transaction is associated with, without revealing which specific transaction in the set is theirs.
- **Purpose:** It allows users to demonstrate compliance with regulatory requirements (like proving funds are not from illicit sources) without revealing their entire transaction history.
- **Functioning:** A user generates a Zero-Knowledge Proof showing that their transaction belongs to a pre-defined set of transactions (the association set). This set can be inclusive (listing transactions that are permitted) or exclusive (listing transactions that are not involved).

A BIT MORE OF ASSOCIATION SETS:

- **Defining the Association Set:** Let S be a set of transactions $\{tx1, tx2, \ldots, txn\}$.
- **Generating a Merkle Tree:** Construct a Merkle tree from S, resulting in a Merkle root R.
- **Creating a Zero-Knowledge Proof:** A user, with a transaction txuser, creates a ZK-SNARK proof P demonstrating that txuser is in S, without revealing which one it is.

  The proof essentially says, "I know a transaction in S that is linked to me, but I won't tell you which one."

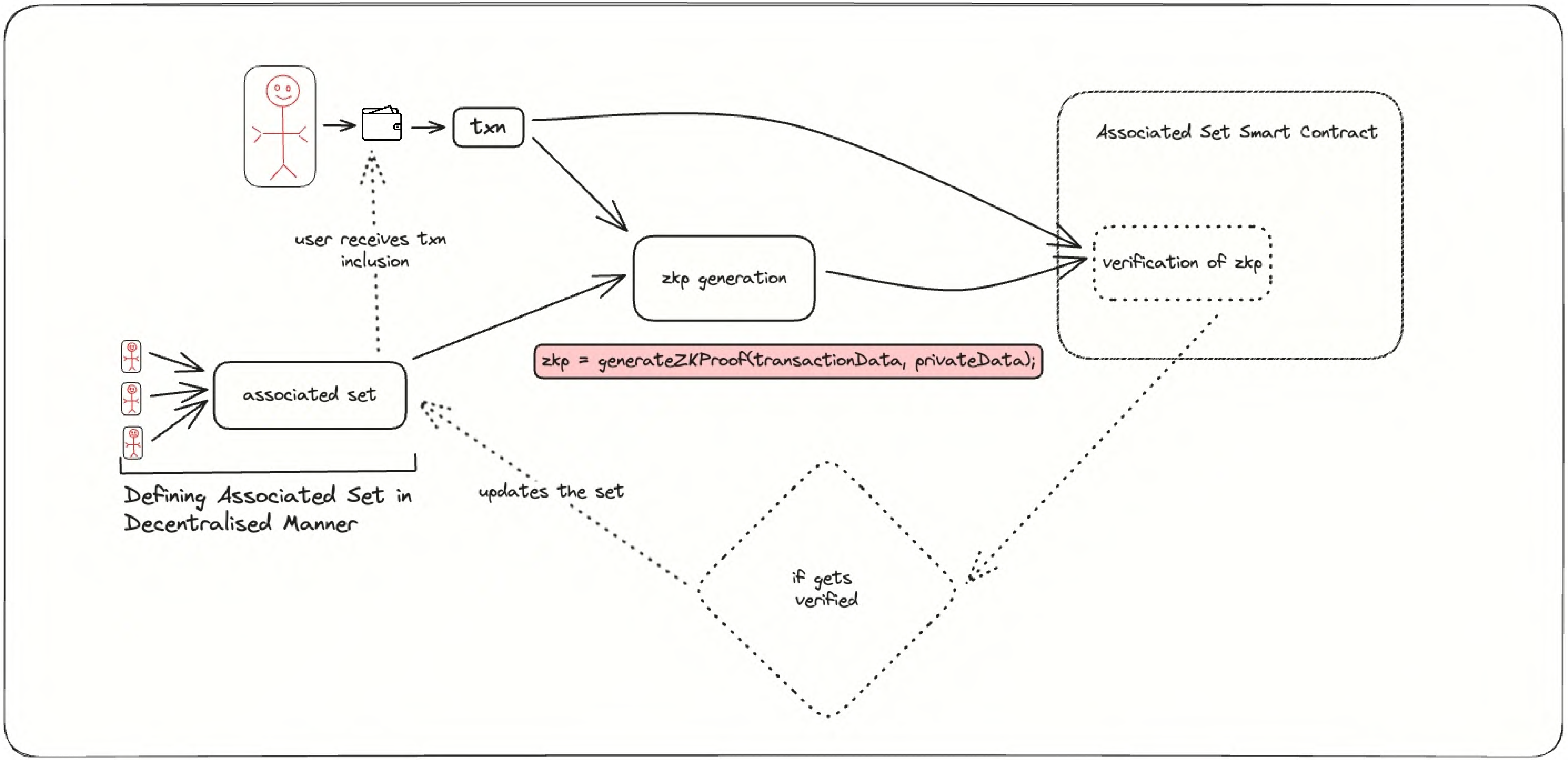
  - Formally, P proves that $txuser \in S$, with the verifier only seeing P and R, but not the specific transaction.

INCLUSION AND EXCLUSION IN ASSOCIATION SETS

- **Inclusion Proof: Selecting transactions that meet compliance standards.**
- **Exclusion Proof: Omitting transactions linked to known illicit activities.**

USER JOURNEY

- **User onboarding and Wallet Integration**
- **Defining Transaction Criteria for Association Set(e.g., transaction amount, source).**
- **Generating Zero-Knowledge Proof (ZKP)**
  const zkp = generateZKProof(transactionData, privateData);
- User submits the transaction along with the ZKP to the ASP.
- ASP verifies the ZKP and updates the association set accordingly.
- User receives a confirmation of the transaction inclusion in the association set.
- For regulatory compliance, the user can request a report of their transactions within the ASP.


txn
Associated Set Smart Contract
user receives txn inclusion
zkp generation
verification of zkp
associated set
zkp = generateZKProof(transactionData, privateData);
Defining Associated Set in Decentralised Manner
updates the set
if gets verified

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FURTHER IMPROVEMENTS W/ ASP:

ASP Provides us with some new and cool features:

- Re-Proofing Mechanism
    - Enable users to regenerate proofs for different association sets or updated criteria without compromising privacy.

- Advance Privacy Features
    - Allowing users to create custom association sets based on real-time criteria, enhancing privacy and compliance adaptability.

TENTATIVE SOLUTION

- An **Associative Set Smart Contract,** managing the state of the ASPs and compatible with the **OLA VM.**
- A **zkSnark Circuit** compatible with the Smart Contract and OLA Vm to generate **Proof of Association** with the ASP.
- **Deployment of the ASP contracts and Circuits on the Ola VM.**

THANK YOU

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